

Java-Codeblooded

E-Commerce Platform

REST API Documentation

Complete Postman Collection Reference

Total APIs: 30 Endpoints
Across 7 Sections

Section	Count	Description
Authentication	2	Register, Login
Categories	5	CRUD Operations
Products	7	CRUD + Search + Filter
Shopping Cart	4	Add, View, Update, Remove
Orders	4	Create, View, Status
Reviews	3	Add, View, Ratings
Users	5	Profile, Password, Delete

Base URL: `http://localhost:8080`

December 10, 2025

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1 Authentication APIs (2 Endpoints)

This section covers user authentication including registration and login functionality.

1.1 API 1: Register New User

POST

Endpoint: `http://localhost:8080/api/auth/register`

Description: Registers a new user account in the system.

Request Body:

```
{
  "email": "raj.shamani@gmail.com",
  "password": "password123",
  "firstName": "Raj",
  "lastName": "Shamani",
  "phone": "9876543210"
}
```

Expected Response (201 Created):

```
{
  "address": null,
  "email": "raj.shamani@gmail.com",
  "firstName": "Raj",
  "id": 12,
  "lastName": "Shamani",
  "name": "Raj Shamani",
  "phone": "9876543210",
  "phoneNumber": "9876543210",
  "role": "CUSTOMER"
}
```

1.2 API 2: Login User

POST

Endpoint: `http://localhost:8080/api/auth/login`

Description: Authenticates a user and returns an access token.

Request Body:

```
{
  "email": "raj.shamani@gmail.com",
  "password": "password123"
}
```

Expected Response (200 OK):

```
{
```

```
    "token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJyYWouc2hhbWFuaUBnbWFpbC5jb20iLCJpYXQiOiE3NjUzNDMxMDAsImV4cCI6MTc2NTQyLjDAuQFVtf0vt9P0BJnkNaODDu7r9cnm7V0CtwsGB7T4",
    "tokenType": "Bearer",
    "expiresAt": "2025-12-11T05:05:00.885819Z",
    "user": {
      "address": null,
      "email": "raj.shamani@gmail.com",
      "firstName": "Raj",
      "id": 12,
      "lastName": "Shamani",
      "name": "Raj Shamani",
      "phone": "9876543210",
      "phoneNumber": "9876543210",
      "role": "CUSTOMER"
    }
  }
```

2 Categories APIs (5 Endpoints)

This section covers category management operations including creation, retrieval, update, and deletion.

2.1 API 3: Get All Categories

GET

Endpoint: `http://localhost:8080/api/categories`

Description: Retrieves all product categories available in the system.

Request Body: None required.

Expected Response (200 OK):

```
[
  {
    "description": "Mobile phones, laptops, and electronic gadgets",
    "id": 1,
    "name": "Electronics",
    "slug": "electronics"
  },
  {
    "description": "Clothing, footwear, and accessories",
    "id": 2,
    "name": "Fashion",
    "slug": "fashion"
  },
  {
    "description": "Home appliances and kitchen essentials",
    "id": 3,
    "name": "Home & Kitchen",
    "slug": "home-kitchen"
  },
  {
    "description": "Fiction, non-fiction, and educational books",
    "id": 4,
    "name": "Books",
    "slug": "books"
  },
  {
    "description": "Sports equipment and fitness accessories",
    "id": 5,
    "name": "Sports & Fitness",
    "slug": "sports-fitness"
  },
  {
    "description": "Fun for everyone",
    "id": 7,
    "name": "Toys & Games",
    "slug": "toys--games"
  },
  {
    "description": "STEEL AND PLASTIC ",
    "id": 9,
```

```
    "name": "BOTTLE",
    "slug": "bottle"
  }
]
```

2.2 API 4: Create Category (Admin)

POST

Endpoint: `http://localhost:8080/api/categories`

Description: Creates a new product category. *Requires Admin privileges.*

Request Body:

```
{
  "name": "Electronics 2",
  "description": "Mobile phones, laptops, and electronic gadgets"
}
```

Expected Response (201 Created):

```
{
  "description": "Mobile phones, laptops, and electronic gadgets",
  "id": 10,
  "name": "Electronics 2",
  "slug": "electronics-2"
}
```

2.3 API 5: Get Category by ID

GET

Endpoint: `http://localhost:8080/api/categories/1`

Description: Retrieves a specific category by its unique identifier.

Path Parameter:

- `id` - Category ID (e.g., 1)

Request Body: None required.

Expected Response (200 OK):

```
{
  "description": "Mobile phones, laptops, and electronic gadgets",
  "id": 1,
  "name": "Electronics",
  "slug": "electronics"
}
```

2.4 API 6: Update Category (Admin)

PUT

Endpoint: `http://localhost:8080/api/categories/1`

Description: Updates an existing category. *Requires Admin privileges.*

Path Parameter:

- `id` - Category ID (e.g., 1)

Request Body:

```
{
  "name": "Electronics & Gadgets",
  "description": "Updated description"
}
```

Expected Response (200 OK):

```
{
  "description": "Updated description",
  "id": 1,
  "name": "Electronics & Gadgets",
  "slug": "electronics--gadgets"
}
```

2.5 API 7: Delete Category (Admin)

DELETE

Endpoint: `http://localhost:8080/api/categories/9`

Description: Deletes a category from the system. *Requires Admin privileges.*

Path Parameter:

- `id` - Category ID (e.g., 9)

Request Body: None required.

Expected Response (200 OK):

```
Category deleted successfully
```


3 Products APIs (7 Endpoints)

This section covers product management including CRUD operations, search, and category filtering.

3.1 API 8: Get All Products (Paginated)

GET

Endpoint: `http://localhost:8080/api/products?page=0&size=10`

Description: Retrieves all products with pagination support.

Query Parameters:

- **page** - Page number (0-indexed)
- **size** - Number of items per page

Request Body: None required.

Expected Response (200 OK):

```
{
  "content": [
    {
      "basePrice": 79900.00,
      "categoryId": 1,
      "categoryName": "Electronics & Gadgets",
      "description": "Apple iPhone 14 with A15 Bionic chip",
      "discount": 5.00,
      "finalPrice": 89567.90,
      "id": 1,
      "imageUrl": "/photos/iphone.jpeg",
      "name": "iPhone 14 128GB",
      "sku": "IPHONE14-128",
      "stock": 97,
      "tax": 18.00
    },
    {
      "basePrice": 16999.00,
      "categoryId": 1,
      "categoryName": "Electronics & Gadgets",
      "description": "Samsung Galaxy M33 with 6GB RAM",
      "discount": 10.01,
      "finalPrice": 18050.93,
      "id": 2,
      "imageUrl": "/photos/galaxy m33.jpeg",
      "name": "Samsung Galaxy M33 5G",
      "sku": "SAMSUNG-M33",
      "stock": 96,
      "tax": 18.00
    },
    {
      "basePrice": 52990.00,
      "categoryId": 1,
      "categoryName": "Electronics & Gadgets",
      "description": "Dell Inspiron 15 with Intel i5",
```

```
    "discount": 8.00,  
    "finalPrice": 57525.94,  
    "id": 3,  
    "imageUrl": "/photos/dell_laptop.jpeg",  
    "name": "Dell Inspiron 15 Laptop",  
    "sku": "DELL-LAPTOP",  
    "stock": 99,  
    "tax": 18.00  
  },  
  {  
    "basePrice": 2999.00,  
    "categoryId": 2,  
    "categoryName": "Fashion",  
    "description": "Levis 511 Slim Fit Jeans",  
    "discount": 20.00,  
    "finalPrice": 2687.10,  
    "id": 4,  
    "imageUrl": "/photos/levis_jeans.jpeg",  
    "name": "Levis 511 Slim Fit Jeans",  
    "sku": "LEVIS-JEANS",  
    "stock": 103,  
    "tax": 12.00  
  },  
  {  
    "basePrice": 5499.00,  
    "categoryId": 2,  
    "categoryName": "Fashion",  
    "description": "Nike Air Max running shoes",  
    "discount": 15.00,  
    "finalPrice": 5235.05,  
    "id": 5,  
    "imageUrl": "/photos/nike_air_max.jpeg",  
    "name": "Nike Air Max Running Shoes",  
    "sku": "NIKE-SHOES",  
    "stock": 100,  
    "tax": 12.00  
  },  
  {  
    "basePrice": 2199.00,  
    "categoryId": 3,  
    "categoryName": "Home & Kitchen",  
    "description": "Prestige Deluxe Alpha",  
    "discount": 12.00,  
    "finalPrice": 2283.44,  
    "id": 6,  
    "imageUrl": "/photos/cooker.jpeg",  
    "name": "Prestige Pressure Cooker 5L",  
    "sku": "PRESTIGE-COOKER",  
    "stock": 100,  
    "tax": 18.00  
  },  
  {  
    "basePrice": 1295.00,  
    "categoryId": 3,  
    "categoryName": "Home & Kitchen",  
    "description": "Philips GC1905 1440-Watt",  
    "discount": 10.00,  
    "finalPrice": 1375.29,
```

```
        "id": 7,
        "imageUrl": "/photos/iron.jpeg",
        "name": "Philips Steam Iron",
        "sku": "PHILIPS-IRON",
        "stock": 100,
        "tax": 18.00
    },
    {
        "basePrice": 299.00,
        "categoryId": 4,
        "categoryName": "Books",
        "description": "Ikigai by H ctor Garc a",
        "discount": 15.00,
        "finalPrice": 254.15,
        "id": 8,
        "imageUrl": "/photos/ikigai.webp",
        "name": "Ikigai: The Japanese Secret",
        "sku": "IKIGAI-BOOK",
        "stock": 100,
        "tax": 0.00
    },
    {
        "basePrice": 499.00,
        "categoryId": 4,
        "categoryName": "Books",
        "description": "Sapiens by Yuval Noah Harari",
        "discount": 20.00,
        "finalPrice": 399.20,
        "id": 9,
        "imageUrl": "/photos/sapiens.webp",
        "name": "Sapiens: A Brief History",
        "sku": "SAPIENS-BOOK",
        "stock": 100,
        "tax": 0.00
    },
    {
        "basePrice": 799.00,
        "categoryId": 5,
        "categoryName": "Sports & Fitness",
        "description": "Premium yoga mat",
        "discount": 25.00,
        "finalPrice": 707.12,
        "id": 10,
        "imageUrl": "/photos/yoga mat.jpeg",
        "name": "Anti-Skid Yoga Mat",
        "sku": "YOGA-MAT",
        "stock": 98,
        "tax": 18.00
    }
],
"empty": false,
"first": true,
"last": false,
"number": 0,
"numberOfElements": 10,
"pageable": {
    "offset": 0,
    "pageNumber": 0,
```

```
    "pageSize": 10,
    "paged": true,
    "sort": {
      "empty": true,
      "sorted": false,
      "unsorted": true
    },
    "unpaged": false
  },
  "size": 10,
  "sort": {
    "empty": true,
    "sorted": false,
    "unsorted": true
  },
  "totalElements": 15,
  "totalPages": 2
}
```

3.2 API 9: Get Product by ID

GET

Endpoint: <http://localhost:8080/api/products/10>

Description: Retrieves a specific product by its unique identifier.

Path Parameter:

- **id** - Product ID (e.g., 10)

Request Body: None required.

Expected Response (200 OK):

```
{
  "basePrice": 799.00,
  "categoryId": 5,
  "categoryName": "Sports & Fitness",
  "description": "Premium yoga mat",
  "discount": 25.00,
  "finalPrice": 707.12,
  "id": 10,
  "imageUrl": "/photos/yoga mat.jpeg",
  "name": "Anti-Skid Yoga Mat",
  "sku": "YOGA-MAT",
  "stock": 98,
  "tax": 18.00
}
```

3.3 API 10: Create Product (Admin)

POST

Endpoint: `http://localhost:8080/api/products`

Description: Creates a new product in the catalog. *Requires Admin privileges.*

Request Body:

```
{
  "name": "iPhone 13 128GB",
  "description": "Apple iPhone 14 with A15 Bionic chip",
  "sku": "IPHONE14-128",
  "basePrice": 79900,
  "tax": 18,
  "discount": 5,
  "stock": 25,
  "categoryId": 17
}
```

Expected Response (201 Created):

```
{
  "id": 1,
  "name": "iPhone 13 128GB",
  "basePrice": 79900.00,
  "finalPrice": 89431.95,
  "stock": 25,
  "categoryName": "Electronics"
}
```

3.4 API 11: Update Product (Admin)

PUT

Endpoint: `http://localhost:8080/api/products/17`

Description: Updates an existing product. *Requires Admin privileges.*

Path Parameter:

- `id` - Product ID (e.g., 17)

Request Body:

```
{
  "name": "iPhone 13 128GB - Updated",
  "basePrice": 74900,
  "stock": 30
}
```

Expected Response (200 OK):

```
{
  "basePrice": 74900,
  "categoryId": 1,
  "categoryName": "Electronics & Gadgets",
}
```

```
{
  "description": "VU 55 inch TV ",
  "discount": 10.00,
  "finalPrice": 79543.80,
  "id": 17,
  "imageUrl": "/photos/television.jpeg",
  "name": "iPhone 13 128GB - Updated",
  "sku": null,
  "stock": 30,
  "tax": 18.00
}
```

3.5 API 12: Delete Product (Admin)

DELETE

Endpoint: `http://localhost:8080/api/products/17`

Description: Deletes a product from the catalog. *Requires Admin privileges.*

Path Parameter:

- `id` - Product ID (e.g., 17)

Request Body: None required.

Expected Response (200 OK):

```
Product deleted successfully
```

3.6 API 13: Search Products

GET

Endpoint: `http://localhost:8080/api/products/search?query=iPhone`

Description: Searches for products matching the query string.

Query Parameter:

- `query` - Search term (e.g., “iPhone”)

Request Body: None required.

Expected Response (200 OK):

```
{
  "content": [
    {
      "basePrice": 79900.00,
      "categoryId": 1,
      "categoryName": "Electronics & Gadgets",
      "description": "Apple iPhone 14 with A15 Bionic chip",
      "discount": 5.00,
      "finalPrice": 89567.90,
      "id": 1,
      "imageUrl": "/photos/iphone.jpeg",

```

```
        "name": "iPhone 14 128GB",
        "sku": "IPHONE14-128",
        "stock": 97,
        "tax": 18.00
    }
],
"empty": false,
"first": true,
"last": true,
"number": 0,
"numberOfElements": 1,
"pageable": {
    "offset": 0,
    "pageNumber": 0,
    "pageSize": 10,
    "paged": true,
    "sort": {
        "empty": true,
        "sorted": false,
        "unsorted": true
    },
    "unpaged": false
},
"size": 10,
"sort": {
    "empty": true,
    "sorted": false,
    "unsorted": true
},
"totalElements": 1,
"totalPages": 1
}
```

3.7 API 14: Get Products by Category

GET

Endpoint: <http://localhost:8080/api/products/category/1>

Description: Retrieves all products belonging to a specific category.

Path Parameter:

- `categoryId` - Category ID (e.g., 1)

Request Body: None required.

Expected Response (200 OK):

```
{
  "content": [
    {
      "basePrice": 79900.00,
      "categoryId": 1,
      "categoryName": "Electronics & Gadgets",
      "description": "Apple iPhone 14 with A15 Bionic chip",
      "discount": 5.00,
```

```

        "finalPrice": 89567.90,
        "id": 1,
        "imageUrl": "/photos/iphone.jpeg",
        "name": "iPhone 14 128GB",
        "sku": "IPHONE14-128",
        "stock": 97,
        "tax": 18.00
    },
    {
        "basePrice": 16999.00,
        "categoryId": 1,
        "categoryName": "Electronics & Gadgets",
        "description": "Samsung Galaxy M33 with 6GB RAM",
        "discount": 10.01,
        "finalPrice": 18050.93,
        "id": 2,
        "imageUrl": "/photos/galaxy m33.jpeg",
        "name": "Samsung Galaxy M33 5G",
        "sku": "SAMSUNG-M33",
        "stock": 96,
        "tax": 18.00
    },
    {
        "basePrice": 52990.00,
        "categoryId": 1,
        "categoryName": "Electronics & Gadgets",
        "description": "Dell Inspiron 15 with Intel i5",
        "discount": 8.00,
        "finalPrice": 57525.94,
        "id": 3,
        "imageUrl": "/photos/dell laptop.jpeg",
        "name": "Dell Inspiron 15 Laptop",
        "sku": "DELL-LAPTOP",
        "stock": 99,
        "tax": 18.00
    },
    {
        "basePrice": 79000.00,
        "categoryId": 1,
        "categoryName": "Electronics & Gadgets",
        "description": "apple 13 inch macbook sky blue",
        "discount": 5.00,
        "finalPrice": 84056.00,
        "id": 14,
        "imageUrl": "/photos/macbook.jpeg",
        "name": "Macbook m4",
        "sku": null,
        "stock": 70,
        "tax": 12.00
    }
],
"empty": false,
"first": true,
"last": true,
"number": 0,
"numberOfElements": 4,
"pageable": {
    "offset": 0,

```



```
    "pageNumber": 0,  
    "pageSize": 10,  
    "paged": true,  
    "sort": {  
      "empty": true,  
      "sorted": false,  
      "unsorted": true  
    },  
    "unpaged": false  
  },  
  "size": 10,  
  "sort": {  
    "empty": true,  
    "sorted": false,  
    "unsorted": true  
  },  
  "totalElements": 4,  
  "totalPages": 1  
}
```

4 Shopping Cart APIs (4 Endpoints)

This section covers shopping cart operations including viewing cart, adding items, updating quantities, and removing items.

4.1 API 15: View Cart

GET

Endpoint: `http://localhost:8080/api/cart?userId=1`

Description: Retrieves the current shopping cart for a user.

Query Parameter:

- `userId` - User ID (e.g., 1)

Request Body: None required.

Expected Response (200 OK):

```
{
  "id": 1,
  "itemCount": 2,
  "items": [
    {
      "id": 1,
      "productId": 1,
      "productImage": "/photos/iphone.jpeg",
      "productName": "iPhone 14 128GB",
      "quantity": 5,
      "subtotal": 447159.75,
      "unitPrice": 89431.95
    },
    {
      "id": 2,
      "productId": 8,
      "productImage": "/photos/ikigai.webp",
      "productName": "Ikigai: The Japanese Secret",
      "quantity": 2,
      "subtotal": 508.30,
      "unitPrice": 254.15
    }
  ],
  "totalAmount": 447668.05
}
```

4.2 API 16: Add Item to Cart

POST

Endpoint: `http://localhost:8080/api/cart/items?userId=8`

Description: Adds a product to the user's shopping cart.

Query Parameter:

- `userId` - User ID (e.g., 8)

Request Body:

```
{
  "productId": 1,
  "quantity": 2
}
```

Expected Response (201 Created):

```
{
  "id": 22,
  "productId": 1,
  "productImage": "/photos/iphone.jpeg",
  "productName": "iPhone 14 128GB",
  "quantity": 2,
  "subtotal": 179135.80,
  "unitPrice": 89567.90
}
```

4.3 API 17: Update Cart Item Quantity

PUT**Endpoint:** `http://localhost:8080/api/cart/items/1?userId=8`**Description:** Updates the quantity of an item in the cart.**Path Parameter:**

- `itemId` - Cart Item ID (e.g., 1)

Query Parameter:

- `userId` - User ID (e.g., 8)

Request Body:

```
{
  "quantity": 3
}
```

Expected Response (200 OK):

```
{
  "id": 22,
  "productId": 1,
  "productImage": "/photos/iphone.jpeg",
  "productName": "iPhone 14 128GB",
  "quantity": 3,
  "subtotal": 268703.70,
  "unitPrice": 89567.90
}
```

4.4 API 18: Remove Item from Cart

DELETE

Endpoint: `http://localhost:8080/api/cart/items/1?userId=8`

Description: Removes an item from the shopping cart.

Path Parameter:

- `itemId` - Cart Item ID (e.g., 1)

Query Parameter:

- `userId` - User ID (e.g., 8)

Request Body: None required.

Expected Response (200 OK):

```
Item removed from cart
```

5 Orders APIs (4 Endpoints)

This section covers order management including checkout, viewing orders, and status updates.

5.1 API 19: Create Order (Checkout)

POST

Endpoint: `http://localhost:8080/api/orders?userId=1`

Description: Creates a new order from the user's cart (checkout).

Query Parameter:

- `userId` - User ID (e.g., 1)

Request Body:

```
{
  "shippingAddress": "Flat 302, MG Road, blr - 411001",
  "paymentMethod": "CARD"
}
```

Payment Methods Available:

- `CARD` - Credit/Debit Card
- `UPI` - UPI Payment
- `COD` - Cash on Delivery
- `NET_BANKING` - Net Banking

Expected Response (201 Created):

```
{
  "createdAt": "2025-12-10T05:08:05.640805Z",
  "id": 6,
  "items": [
    {
      "id": 11,
      "productId": 1,
      "productName": "iPhone 14 128GB",
      "quantity": 5,
      "subtotal": 447159.75,
      "unitPrice": 89431.95
    },
    {
      "id": 12,
      "productId": 8,
      "productName": "Ikigai: The Japanese Secret",
      "quantity": 2,
      "subtotal": 508.30,
      "unitPrice": 254.15
    }
  ],
}
```

```
{
  "orderNumber": "ORD-20251210103805",
  "paymentMethod": "CARD",
  "paymentStatus": "SUCCESS",
  "shippingAddress": "Flat 302, MG Road, blr - 411001",
  "status": "CONFIRMED",
  "totalAmount": 447668.05
}
```

5.2 API 20: Get User Orders

GET

Endpoint: `http://localhost:8080/api/orders?userId=1`

Description: Retrieves all orders for a specific user.

Query Parameter:

- `userId` - User ID (e.g., 1)

Request Body: None required.

Expected Response (200 OK):

```
{
  "content": [
    {
      "createdAt": "2025-12-10T05:08:05.640805Z",
      "id": 6,
      "items": [
        {
          "id": 11,
          "productId": 1,
          "productName": "iPhone 14 128GB",
          "quantity": 5,
          "subtotal": 447159.75,
          "unitPrice": 89431.95
        },
        {
          "id": 12,
          "productId": 8,
          "productName": "Ikigai: The Japanese Secret",
          "quantity": 2,
          "subtotal": 508.30,
          "unitPrice": 254.15
        }
      ],
      "orderNumber": "ORD-20251210103805",
      "paymentMethod": "CARD",
      "paymentStatus": "SUCCESS",
      "shippingAddress": "Flat 302, MG Road, blr - 411001",
      "status": "CONFIRMED",
      "totalAmount": 447668.05
    }
  ],
  "empty": false,
  "first": true,
}
```

```
{
  "last": true,
  "number": 0,
  "numberOfElements": 1,
  "pageable": {
    "offset": 0,
    "pageNumber": 0,
    "pageSize": 10,
    "paged": true,
    "sort": {
      "empty": true,
      "sorted": false,
      "unsorted": true
    },
    "unpaged": false
  },
  "size": 10,
  "sort": {
    "empty": true,
    "sorted": false,
    "unsorted": true
  },
  "totalElements": 1,
  "totalPages": 1
}
```

5.3 API 21: Get Order by ID

GET

Endpoint: <http://localhost:8080/api/orders/1?userId=1>

Description: Retrieves details of a specific order.

Path Parameter:

- `orderId` - Order ID (e.g., 1)

Query Parameter:

- `userId` - User ID (e.g., 1)

Request Body: None required.

Expected Response (200 OK):

```
{
  "id": 1,
  "orderNumber": "ORD-20241210-001",
  "status": "PENDING",
  "totalAmount": 178962.90,
  "shippingAddress": "Flat 302, MG Road, Pune - 411001",
  "paymentMethod": "CARD",
  "paymentStatus": "PENDING",
  "items": [...],
  "createdAt": "2024-12-10T10:00:00"
}
```

5.4 API 22: Update Order Status (Admin)

PUT

Endpoint: `http://localhost:8080/api/orders/3/status`

Description: Updates the status of an order. *Requires Admin privileges.*

Path Parameter:

- `orderId` - Order ID (e.g., 1)

Request Body:

```
{
  "status": "SHIPPED"
}
```

Order Status Flow:

PENDING → CONFIRMED → SHIPPED → DELIVERED

(Can also be set to *CANCELLED* at any point)

Expected Response (200 OK):

```
{
  "createdAt": "2025-12-07T09:53:58.835850Z",
  "id": 3,
  "items": [
    {
      "id": 3,
      "productId": 1,
      "productName": "iPhone 14 128GB",
      "quantity": 2,
      "subtotal": 179135.80,
      "unitPrice": 89567.90
    },
    {
      "id": 4,
      "productId": 2,
      "productName": "Samsung Galaxy M33 5G",
      "quantity": 1,
      "subtotal": 18052.94,
      "unitPrice": 18052.94
    },
    {
      "id": 5,
      "productId": 3,
      "productName": "Dell Inspiron 15 Laptop",
      "quantity": 1,
      "subtotal": 57525.94,
      "unitPrice": 57525.94
    }
  ],
  "orderNumber": "ORD-20251207152358",
  "paymentMethod": "COD",
  "paymentStatus": "SUCCESS",
  "shippingAddress": "BLR",
}
```



```
"status": "SHIPPED",  
"totalAmount": 254714.68  
}
```

6 Reviews APIs (3 Endpoints)

This section covers product review functionality including viewing, adding reviews, and average ratings.

6.1 API 23: Get Product Reviews

GET

Endpoint: `http://localhost:8080/api/products/1/reviews`

Description: Retrieves all reviews for a specific product.

Path Parameter:

- `productId` - Product ID (e.g., 1)

Request Body: None required.

Expected Response (200 OK):

```
[
  {
    "comment": "Great product!",
    "createdAt": "2025-12-07T09:17:17.880937Z",
    "id": 2,
    "productId": 1,
    "productName": "iPhone 14 128GB",
    "rating": 5,
    "userId": 7,
    "userName": "Test Admin"
  }
]
```

6.2 API 24: Add Review

POST

Endpoint: `http://localhost:8080/api/products/11/reviews?userId=4`

Description: Adds a new review for a product.

Path Parameter:

- `productId` - Product ID (e.g., 11)

Query Parameter:

- `userId` - User ID (e.g., 4)

Request Body:

```
{
  "productId": 1,
  "rating": 5,
  "comment": "Amazing product! Highly recommended."
}
```

Rating Scale: 1-5 stars

Expected Response (201 Created):

```
{
  "comment": "Amazing product! Highly recommended.",
  "createdAt": "2025-12-10T05:09:12.767098Z",
  "id": 5,
  "productId": 3,
  "productName": "Dell Inspiron 15 Laptop",
  "rating": 5,
  "userId": 4,
  "userName": "Sneha Singh"
}
```

6.3 API 25: Get Average Rating

GET

Endpoint: <http://localhost:8080/api/products/1/reviews/average>

Description: Retrieves the average rating for a specific product.

Path Parameter:

- productId - Product ID (e.g., 1)

Request Body: None required.

Expected Response (200 OK):

```
5.0
```

7 Users APIs (5 Endpoints)

This section covers user profile management including viewing, updating profiles, password changes, and account deletion.

7.1 API 26: Get User Profile

GET

Endpoint: `http://localhost:8080/api/users/me?userId=1`

Description: Retrieves the profile of the authenticated user.

Query Parameter:

- `userId` - User ID (e.g., 1)

Request Body: None required.

Expected Response (200 OK):

```
{
  "address": null,
  "email": "raj.sharma@gmail.com",
  "firstName": "Raj",
  "id": 1,
  "lastName": "Sharma",
  "name": "Raj Sharma",
  "phone": "9876543210",
  "phoneNumber": "9876543210",
  "role": "CUSTOMER"
}
```

7.2 API 27: Update Profile

PUT

Endpoint: `http://localhost:8080/api/users/me?userId=1`

Description: Updates the user's profile information.

Query Parameter:

- `userId` - User ID (e.g., 1)

Request Body:

```
{
  "firstName": "Raj",
  "lastName": "Sharma",
  "phone": "9876543210"
}
```

Expected Response (200 OK):

```
{
  "address": null,
  "email": "raj.sharma@gmail.com",
  "firstName": "Raj",
  "id": 1,
  "lastName": "Sharma",
  "name": "Raj Sharma",
  "phone": "9876543210",
  "phoneNumber": "9876543210",
  "role": "CUSTOMER"
}
```

7.3 API 28: Get All Users (Admin)

GET

Endpoint: <http://localhost:8080/api/users>

Description: Retrieves all registered users. *Requires Admin privileges.*

Request Body: None required.

Expected Response (200 OK):

```
[
  {
    "address": null,
    "email": "raj.sharma@gmail.com",
    "firstName": "Raj",
    "id": 1,
    "lastName": "Sharma",
    "name": "Raj Sharma",
    "phone": "9876543210",
    "phoneNumber": "9876543210",
    "role": "CUSTOMER"
  },
  {
    "address": null,
    "email": "amit.kumar@gmail.com",
    "firstName": "Amit",
    "id": 3,
    "lastName": "Kumar",
    "name": "Amit Kumar",
    "phone": "9876543212",
    "phoneNumber": "9876543212",
    "role": "CUSTOMER"
  },
  {
    "address": null,
    "email": "sneha.singh@gmail.com",
    "firstName": "Sneha",
    "id": 4,
    "lastName": "Singh",
    "name": "Sneha Singh",
    "phone": "9876543213",
    "phoneNumber": "9876543213",
    "role": "CUSTOMER"
  }
]
```

```
    "role": "CUSTOMER"
  },
  {
    "address": null,
    "email": "raj.sharma1@gmail.com",
    "firstName": "Raj",
    "id": 5,
    "lastName": "Sharma",
    "name": "Raj Sharma",
    "phone": "9876543210",
    "phoneNumber": "9876543210",
    "role": "CUSTOMER"
  },
  {
    "address": null,
    "email": "priya.patel1@gmail.com",
    "firstName": "Priya",
    "id": 6,
    "lastName": "Patel",
    "name": "Priya Patel",
    "phone": "9876543211",
    "phoneNumber": "9876543211",
    "role": "CUSTOMER"
  },
  {
    "address": null,
    "email": "test123@gmail.com",
    "firstName": "Test",
    "id": 7,
    "lastName": "Admin",
    "name": "Test Admin",
    "phone": "1234567890",
    "phoneNumber": "1234567890",
    "role": "ADMIN"
  },
  {
    "address": null,
    "email": "test@example.com",
    "firstName": "Test",
    "id": 8,
    "lastName": "User",
    "name": "Test User",
    "phone": "09876543211234567890",
    "phoneNumber": "09876543211234567890",
    "role": "CUSTOMER"
  },
  {
    "address": null,
    "email": "testapi@test.com",
    "firstName": "API",
    "id": 9,
    "lastName": "Test",
    "name": "API Test",
    "phone": null,
    "phoneNumber": null,
    "role": "CUSTOMER"
  },
  {
```

```
    "address": null,
    "email": "test1234@gmail.com",
    "firstName": "Aryan",
    "id": 10,
    "lastName": "",
    "name": "Aryan ",
    "phone": "1234567890",
    "phoneNumber": "1234567890",
    "role": "CUSTOMER"
  },
  {
    "address": null,
    "email": "testing@gmail.com",
    "firstName": "Test",
    "id": 11,
    "lastName": "",
    "name": "Test ",
    "phone": "0987654321",
    "phoneNumber": "0987654321",
    "role": "CUSTOMER"
  },
  {
    "address": null,
    "email": "raj.shamani@gmail.com",
    "firstName": "Raj",
    "id": 12,
    "lastName": "Shamani",
    "name": "Raj Shamani",
    "phone": "9876543210",
    "phoneNumber": "9876543210",
    "role": "CUSTOMER"
  }
]
```

7.4 API 29: Change Password

PUT

Endpoint: <http://localhost:8080/api/users/1/password>

Description: Changes the user's password.

Path Parameter:

- **userId** - User ID (e.g., 1)

Request Body:

```
{
  "currentPassword": "password123",
  "newPassword": "newpassword456"
}
```

Expected Response (200 OK):

Password changed successfully

7.5 API 30: Delete User

DELETE

Endpoint: `http://localhost:8080/api/users/1`

Description: Deletes a user account from the system.

Path Parameter:

- `userId` - User ID (e.g., 1)

Request Body: None required.

Expected Response (200 OK):

```
User deleted successfully
```


8 Appendix: HTTP Response Codes

Table 1: Standard HTTP Response Codes

Code	Status	Description
200	OK	Successful GET, PUT, or DELETE request
201	Created	Successful POST request (resource created)
400	Bad Request	Missing request body or validation error
401	Unauthorized	Invalid or missing authentication token
403	Forbidden	Insufficient permissions (Admin-only endpoint)
404	Not Found	Requested resource does not exist
500	Internal Server Error	Server-side error occurred

9 Appendix: Complete API Summary

No.	API Name	Method	Endpoint
1	Register New User	POST	/api/auth/register
2	Login User	POST	/api/auth/login
3	Get All Categories	GET	/api/categories
4	Create Category	POST	/api/categories
5	Get Category by ID	GET	/api/categories/{id}
6	Update Category	PUT	/api/categories/{id}
7	Delete Category	DELETE	/api/categories/{id}
8	Get All Products	GET	/api/products
9	Get Product by ID	GET	/api/products/{id}
10	Create Product	POST	/api/products
11	Update Product	PUT	/api/products/{id}
12	Delete Product	DELETE	/api/products/{id}
13	Search Products	GET	/api/products/search
14	Products by Category	GET	/api/products/category/{id}
15	View Cart	GET	/api/cart
16	Add to Cart	POST	/api/cart/items
17	Update Cart Item	PUT	/api/cart/items/{id}
18	Remove from Cart	DELETE	/api/cart/items/{id}
19	Create Order	POST	/api/orders
20	Get User Orders	GET	/api/orders
21	Get Order by ID	GET	/api/orders/{id}
22	Update Order Status	PUT	/api/orders/{id}/status
23	Get Product Reviews	GET	/api/products/{id}/reviews
24	Add Review	POST	/api/products/{id}/reviews
25	Get Average Rating	GET	/api/products/{id}/reviews/average
26	Get User Profile	GET	/api/users/me
27	Update Profile	PUT	/api/users/me

No.	API Name	Method	Endpoint
28	Get All Users	GET	/api/users
29	Change Password	PUT	/api/users/{id}/password
30	Delete User	DELETE	/api/users/{id}

10 Appendix: API Method Distribution

Table 3: HTTP Methods Distribution

HTTP Method	Count	Percentage
GET	14	46.7%
POST	8	26.7%
PUT	5	16.7%
DELETE	3	10.0%
Total	30	100%

11 Appendix: Postman Testing Tips

11.1 Setting Up Requests

1. For POST/PUT requests:

- Go to the **Body** tab
- Select **raw**
- Choose **JSON** from the dropdown
- Paste the JSON body

2. Replace placeholders:

- `userId=1` → Use actual user ID from registration
- `/products/1` → Use actual product ID

11.2 Recommended Test Flow

1. Register a new user (obtain `userId`)
2. Login with the registered credentials
3. Create categories (Admin)
4. Create products (Admin)
5. Add items to cart
6. Complete checkout (create order)
7. Add product reviews
8. View orders and order details